

Prisma Setup Troubleshooting Guide

Common Issues and Solutions

Issue 1: EPERM Errors During Setup

Symptom:

- ✗ File locking error detected (EPERM)
- ✗ Max retries reached. Prisma generate failed.

What This Means: Another process is holding locks on Prisma DLL files in `node_modules/.prisma/client/`, preventing regeneration.

Solution Steps:

Step 1: Try Standard Retry (Already Automatic)

The `prisma-generate-safe.js` script automatically:

- Detects EPERM errors
- Cleans up temp files
- Unlocks Prisma files
- Retries up to 3 times

If this succeeds, no action needed.

Step 2: Manual Nuclear Cleanup (If Retries Fail)

⚠ **IMPORTANT:** Only use nuclear cleanup when:

- Standard retries have failed
- No production services are running
- You're prepared for ALL Node processes to be killed

Run Nuclear Cleanup:

```
# This will kill ALL Node.js processes and remove .prisma directory
node scripts/prisma-nuclear-cleanup.js

# Then retry setup
npm run setup
```

Step 3: Production Server (Services Running)

⚠ **DO NOT** use nuclear cleanup on production with running services!

Instead:

```
# 1. Stop all services first
npm run sync-service:stop
sc.exe stop "electricity-tokens.exe"

# 2. Run safe setup
npm run setup:production-safe

# 3. Restart services after
npm run sync-service:start
sc.exe start "electricity-tokens.exe"
```

Issue 2: Setup Stops After Prisma Generation

Symptom: Setup completes Prisma generation but doesn't continue with migrations, seeding, or building.

Cause: You may have manually run setup with `--nuclear` flag, which kills the parent setup process.

Solution:

```
# Use standard setup (no --nuclear flag)
npm run setup

# Only use nuclear cleanup manually if standard retry fails:
# 1. Stop the failed setup (Ctrl+C)
# 2. Run: node scripts/prisma-nuclear-cleanup.js
# 3. Run setup again: npm run setup
```

Issue 3: Services Won't Stop

Symptom:

```
sc.exe stop "electricity-tokens.exe"
# Returns error or service won't stop
```

Solutions:

Option 1: Force Stop

```
sc.exe stop "electricity-tokens.exe" /force
```

Option 2: Task Manager

1. Open Task Manager

2. Find "electricity-tokens.exe" or "node.exe" processes
3. End the process
4. Verify with: `sc.exe query "electricity-tokens.exe"`

Option 3: Restart Server

```
# Last resort if services are stuck
shutdown /r /t 0
```

Issue 4: "Nuclear Cleanup" Killed My Setup Script

Why This Happens: Nuclear cleanup kills ALL Node.js processes except itself. If setup script calls nuclear cleanup, setup gets killed.

Solution: Don't use `--nuclear` flag in setup scripts. Use it only as a standalone command:

```
# ❌ WRONG - This will kill the setup process
node scripts/setup-fresh-install.js --nuclear

# ✅ CORRECT - Let setup use standard retry logic
npm run setup

# If setup fails with EPERM after retries, THEN run nuclear cleanup separately:
node scripts/prisma-nuclear-cleanup.js
npm run setup
```

Recommended Workflow

Development Machine (No Services Running)

```
# Standard setup with automatic retry
npm run setup

# If that fails with EPERM after 3 retries:
node scripts/prisma-nuclear-cleanup.js
npm run setup
```

Production Server (Services Running)

```
# 1. Stop services
npm run sync-service:stop
sc.exe stop "electricity-tokens.exe"

# 2. Verify services stopped
npm run sync-service:status
```

```
sc.exe query "electricity-tokens.exe"

# 3. Run production-safe setup
npm run setup:production-safe

# 4. Restart services
npm run sync-service:start
sc.exe start "electricity-tokens.exe"

# 5. Verify services running
npm run sync-service:status
sc.exe query "electricity-tokens.exe"
```

Prevention Tips

1. Always Stop Services Before Setup

On production servers, always stop services that use Prisma before running setup.

2. Don't Run Multiple Setups Simultaneously

Only run one setup process at a time. Multiple concurrent setups can cause lock conflicts.

3. Close Development Servers

Before running setup, stop any running dev servers:

```
# Stop dev server (Ctrl+C in the terminal running it)
# Or find and kill the process in Task Manager
```

4. Avoid Manual Prisma Commands During Setup

Let setup scripts handle Prisma operations. Don't run manual `prisma generate` during setup.

5. Use Production-Safe Script on Servers

Always use `npm run setup:production-safe` on production servers - it checks for running services.

Quick Reference

Scenario	Command
Fresh setup (dev, no services)	<code>npm run setup</code>
Fresh setup (production, with services)	<code>npm run setup:production-safe</code>
Manual nuclear cleanup	<code>node scripts/prisma-nuclear-cleanup.js</code>
Check locked files	<code>node scripts/cleanup-prisma-locks.js --manual</code>

Scenario	Command
Stop sync service	<code>npm run sync-service:stop</code>
Stop electricity service	<code>sc.exe stop "electricity-tokens.exe"</code>
Start sync service	<code>npm run sync-service:start</code>
Start electricity service	<code>sc.exe start "electricity-tokens.exe"</code>

When to Contact Support

Contact support if:

- EPERM errors persist after nuclear cleanup
- Services won't stop even with force
- Setup consistently fails on a specific step
- You see database connection errors
- Migrations fail repeatedly

Additional Resources

- [ADMIN-INSTALLATION-GUIDE.md](#) - Production deployment guide
- [prisma-windows-locking-fix.md](#) - Technical details on Windows locking